Zhu Xu BMI 500 Homework 7 : HW 2

Hypotheses

(i) Reducing movement is expected to lower the infection peak and its timing. Agents are less likely to encounter new infected ones, but agents shares the same cell can still infect each other.

(ii) Individual simulation runs are unlikely to output similar results. In some cases, the outbreak can spread quickly, meanwhile, the five agent who was infected initially could die early.

(iii) I think the average agent-based curves are expected to resemble the SIR model. But there could also be some differences because the infection in the agent-based model depends on the agents' positions and random movements.

(iv) I think the effect of population density can be explored. The grid size can be changed, and the population can be fixed at 200 people. Higher density is expected to increase the infection peak because more agents are sharing the same cell. Also, different movement probabilities will be tested.